AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Gene expression and genetics

This page follows a gene from code to consequence: how the genetic code is read, how transcription and translation build a protein, how mutations change it, how alleles are inherited, and how cystic fibrosis and genetic screening put these ideas to use.

SPEC 2.11 The nature of the genetic code

WHAT THIS SPEC POINT ASKS

Understand the nature of the genetic code: triplet, non-overlapping and degenerate.

- **Triplet**: a sequence of **three bases** (a **codon** on mRNA) codes for **one amino acid**. 4^3 = **64** possible triplets, which is plenty for 20 amino acids (a doublet code would give only 4^2 = 16).
- **Non-overlapping**: each base is read **once**, as part of **only one** triplet.
- **Degenerate**: most amino acids are coded for by **more than one** triplet (e.g. glycine by GGU, GGC, GGA, GGG). This means some substitution mutations change no amino acid (2.14).
- Some triplets are **start** (AUG) or **stop** (UAA, UAG, UGA) signals.
- (The code is also **universal**: the same triplets code for the same amino acids in almost all organisms.)

WORKED EXAMPLE

Non-overlapping vs overlapping

mRNA: A U G G C A U U C

Read **non-overlapping**: AUG · GCA · UUC → 3 codons → 3 amino acids.

If it overlapped, AUG, UGG, GGC, GCA… would all be read, so one base change would alter several amino acids.

MARK-SCHEME WORDING

Non-overlapping: "a base in one triplet **does not code for any other** amino acid" · Degenerate: "**most amino acids are coded for by more than one codon**" (accept: only Met and Trp have one codon) · "a sequence of 12 bases codes for **4** amino acids, because each amino acid needs 3 bases" · Universal: "the **same triplet codes for the same amino acid in (almost) all organisms**"

Triplet: "**three bases code for one amino acid**", with an example from the table (e.g. AAA = Lys). Each feature needs a table example, and don't define a term with the term itself.

Why a triplet: "two bases would only give **16** (4^2) combinations, not enough for **20** amino acids" (plus start and stop); three give 64 (4^3).

Degenerate also "**minimises the effect of mutations**".

Stop codons: "**no more amino acids can be added**". 3 of the 64 codons are stop codons, so 61 ÷ 64 = **95.31%** code for amino acids.

Often asked as: describe each feature using a codon table (6-mark); why a triplet and not a doublet; count bases, codons or amino acids; explain why a substitution had no effect.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The start codon **AUG also codes for methionine**, so Met appears inside polypeptides as well as at the start (WBI11 Jun 2024).
- Reading a **codon wheel**: start at the centre with the first base and work outwards (WBI11 Oct 2025).

COMMON MISTAKE

Check whether a table is written as **mRNA codons** or **DNA triplets**. A DNA template triplet **CAG** gives the mRNA codon **GUC**. Always convert to mRNA first before using a codon table.

Swapping **degenerate** (several codons for one amino acid) and **non-overlapping** (each base read once). It's the **code** that is degenerate, not DNA or the bases.

SPEC 2.12 What a gene is

WHAT THIS SPEC POINT ASKS

Know that a gene is a sequence of bases on a DNA molecule that codes for a sequence of amino acids in a polypeptide chain.

- A **gene** is a **sequence of bases on DNA** that codes for the **sequence of amino acids** in a **polypeptide**.
- Its position on a chromosome is its **locus**. One DNA molecule (chromosome) carries many genes.
- The chain of cause and effect: **base sequence** → **amino acid sequence (primary structure)** → **3-D shape** → **function** of the protein.

WORKED EXAMPLE

Counting

A polypeptide of 150 amino acids needs at least 150 × 3 = **450 bases** in the coding part of the gene (plus 3 for the stop codon).

MARK-SCHEME WORDING

Gene: "a **sequence of (DNA) bases** that codes for a **(poly)peptide**" (accept "primary structure").

Allele: "a **version / form of a gene**". Gene v allele questions need **named examples** from the context, or they score 1.

mRNA "carries a **copy of the gene** to the ribosomes". "mRNA copies DNA" or "a copy of DNA" is too vague.

COMMON MISTAKE

Mixing up gene and allele. The gene is the section of DNA for one polypeptide (e.g. the CFTR gene); alleles are its versions (e.g. the normal and the faulty CFTR allele).

SPEC 2.13 Protein synthesis: transcription and translation

WHAT THIS SPEC POINT ASKS

(i) Understand transcription and translation, including RNA polymerase, mRNA, tRNA, ribosomes, and start and stop codons. (ii) Understand the roles of the DNA template (antisense) strand, codons on mRNA and anticodons on tRNA.

Transcription (in the nucleus)

1. The DNA double helix **unwinds** and **hydrogen bonds break** in the region of the gene, separating the strands.
2. Only **one strand**, the **template (antisense) strand**, is used.
3. **Free RNA nucleotides** pair with the exposed DNA bases by **complementary base pairing** (A–U, T–A, C–G, G–C).
4. **RNA polymerase** joins the RNA nucleotides with **phosphodiester bonds**, forming **mRNA**.
5. mRNA **detaches**; the DNA re-forms its helix. mRNA leaves the nucleus through a **nuclear pore**.

The mRNA has the **same sequence as the coding (sense) strand**, except **U replaces T**.

Translation (on ribosomes in the cytoplasm)

1. The mRNA attaches to a **ribosome**.
2. **tRNA** molecules each carry a **specific amino acid** and have an **anticodon** (three bases).
3. The tRNA whose anticodon is **complementary** to the **start codon** (AUG) binds first.
4. The next tRNA binds to the next codon. The ribosome holds **two tRNAs** at a time.
5. A **peptide bond** forms between the two amino acids (**condensation**), using ATP.
6. The ribosome **moves along one codon**; the first tRNA leaves (to collect another amino acid).
7. This repeats until a **stop codon**, which has no tRNA; the **polypeptide is released**.

Worked conversion (a very common question)

WORKED EXAMPLE

DNA → mRNA → tRNA → amino acids

	Triplet 1	Triplet 2	Triplet 3
DNA template (antisense) strand	TAC	GGT	ACT
mRNA codon	AUG	CCA	UGA
tRNA anticodon	UAC	GGU	—
Amino acid	Met (start)	Pro	stop

The **coding (sense)** strand would read ATG CCA TGA, the same as mRNA with T for U.

Shortcut: the **anticodon** is the template triplet with U in place of T (TAC → UAC).

mRNA vs tRNA

	mRNA	tRNA
Shape	single, linear strand	single strand folded into a **clover-leaf**, held by H-bonds
Key feature	**codons**	**anticodon** at one end; **amino acid binding site** at the other
Job	carries the gene's code to the ribosome	brings the specific amino acid to the ribosome

MARK-SCHEME WORDING

Exact names: "**transcription**", "**translation**", "**RNA polymerase**" (not DNA polymerase, helicase or just "polymerase"). Hedging between two names scores 0.

mRNA: "carries a copy of the gene out of the nucleus to the ribosomes"; its codons set the order of the amino acids.

tRNA: "carries a **specific** amino acid"; anticodon "**complementary** to the codon"; "holds the amino acid in place until the peptide bond forms".

Ribosome: site of translation; "**holds (two) tRNAs** on the mRNA" while the peptide bond forms.

Start codon: "starts translation" and codes for **methionine**. Stop codons have **no tRNA**.

Effect of a mutation: "the change is **copied into the mRNA**" → a different **codon** → "a tRNA carrying a different amino acid binds" → it's joined into the polypeptide by a peptide bond.

Often asked as: DNA → mRNA → anticodon → amino acids from a table; roles of mRNA, tRNA, RNA polymerase or ribosomes; compare start and stop codons.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **RNA polymerase binds to the promoter** (a base sequence just before the gene), separates the strands and forms phosphodiester bonds (WBI11 Jun 2024).
- **Start v stop codons**: 1 start codon (AUG, methionine) and 3 stop codons (UAA, UAG, UGA). No tRNA has an anticodon for a stop codon, so the polypeptide is released (WBI11 Jun 2024).
- **mRNA medicines**: lipid nanoparticles protect the mRNA from hydrolytic enzymes and fuse with the cell surface membrane; the mRNA is then translated (WBI11 Oct 2019).
- **RNA interference and antisense drugs** stop the mRNA being **translated**; transcription carries on (WBI11 Jan 2025).
- **Codominance** at the molecular level: both alleles are transcribed and translated, so both proteins are made (WBI11 Jan 2023).

COMMON MISTAKE

Mixing up **DNA polymerase** (replication) with **RNA polymerase** (transcription). Writing **T** in mRNA or tRNA (it's **U**). The **anticodon is on tRNA**, the **codon on mRNA**.

Saying peptide bonds form in transcription: they form in **translation**.

Thinking stop codons have a tRNA. Giving the mRNA sequence when the question asks for the amino acids.

Naming an enzyme or molecule when the question asks for its **role**.

SPEC 2.14 Mutations

WHAT THIS SPEC POINT ASKS

(i) Understand how errors in DNA replication cause mutations (substitution, insertion, deletion). (ii) Know that some mutations cause cancer or genetic disorders, but many have no observable effect.

A **gene mutation** is a change in the **base sequence** of DNA, usually from an error during **replication**.

Type	What happens	Effect
Substitution	one base replaced by another	changes **one codon** only: may give the **same** amino acid (degenerate code: *silent*), **one different** amino acid, or a **stop codon** (shortened protein)
Insertion	an extra base added	**frameshift**: every codon after it is read differently → many amino acids change; often a stop codon appears
Deletion	a base lost	**frameshift**, as for insertion

WORKED EXAMPLE

Worked example

Original mRNA: AUG **G**CA UUC → Met – Ala – Phe

Substitution (G→C): AUG **C**CA UUC → Met – **Pro** – Phe (one amino acid changes)

Deletion of that G: AUG CAU UC… → Met – **His** – … (everything after the mutation changes)

Why a mutation may or may not matter

- **No effect**: the new codon codes for the **same** amino acid; or the change is in non-coding DNA; or the new amino acid has similar properties and doesn't change the shape.
- **Harmful**: a different amino acid changes the **primary structure** → different bonds between R groups → different **tertiary structure** → the protein doesn't function (e.g. an enzyme's active site no longer complementary). Examples: **cystic fibrosis** (2.16), sickle cell anaemia.
- **Cancer**: mutations in genes that control **cell division** can lead to uncontrolled division and a **tumour**.
- Mutations in **gametes** can be **inherited**; mutations in body cells are not.

EXAM TIP

Explain the chain fully: base sequence → **codon/triplet** → **amino acid sequence (primary structure)** →

bonds / 3-D (tertiary) shape → function.

MARK-SCHEME WORDING

Mutation: "a change in the **base sequence** of DNA".

Use the spec's names: **substitution, insertion, deletion**. "Point", "frameshift", "addition", "nonsense" and "duplication" are **ignored** as type names.

Substitution: affects **one triplet**; may be **silent** (degenerate code, or a non-coding region); may change **one amino acid** → different bonds between R groups → different structure and activity; may make a **stop codon** → shorter polypeptide, or none.

Insertion / deletion: shifts the reading frame, so **every later amino acid** may change; the **earlier** in the gene, the bigger the effect.

Cancer: "cells **dividing out of control**" as a result of mutation; they may spread.

Often asked as: name the mutation type from two sequences; use a codon table to explain the effect on the protein (6-mark); explain why a mutation had no effect.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Chromosome mutations** appear in school assessments and some data questions: whole chromosomes gained or lost through **non-disjunction** in meiosis (e.g. **Down's syndrome**, trisomy 21: three copies of chromosome 21), and changes to chromosome structure (deletion, duplication, inversion, translocation of a section). These change many genes at once.
- **Why mutation rates differ** (WBI11 Oct 2024): faster cell division (more replication), less **DNA repair or proofreading**, more exposure to **mutagens** such as UV, more DNA. Mutations also **build up with age**.
- **Proto-oncogenes**: normal genes that control cell division; mutated, they can cause cancer (WBI11 Jan 2025).

COMMON MISTAKE

"Substitution of bases" just repeats the name: say **which** base changed and what that does to the codon and the amino acid.

Generic answers: apply the mutation to the **table or sequence given** and link it to the protein's **function**.

SPEC 2.15 Inheritance: terms, crosses, pedigrees, sex linkage

WHAT THIS SPEC POINT ASKS

(i) Understand gene, allele, genotype, phenotype, recessive, dominant, codominance, homozygote, heterozygote. (ii) Understand monohybrid inheritance, including interpreting pedigree diagrams. (iii) Understand sex linkage on the X chromosome, including red-green colour blindness.

Key terms

Term	Meaning
Gene	a sequence of bases on DNA coding for a polypeptide; found at a **locus**
Allele	a different **version** of the same gene
Genotype	the **alleles** an organism has (e.g. Ff)
Phenotype	the **observable characteristics**, from genotype + environment
Dominant	allele always expressed when present (capital letter)
Recessive	allele expressed only when **no dominant allele** is present (homozygous)
Codominance	**both** alleles expressed in the heterozygote (e.g. roan cattle, blood group AB)
Homozygote	two **identical** alleles (FF or ff)
Heterozygote	two **different** alleles (Ff); if one is a harmful recessive, the person is a **carrier**

Monohybrid crosses

WORKED EXAMPLE

Two heterozygous parents (dominant / recessive)

Parents: **Ff × Ff** · Gametes: F, f and F, f

	F	f
F	FF	Ff
f	Ff	ff

Genotypes 1 FF : 2 Ff : 1 ff · Phenotypes **3 dominant : 1 recessive**. Probability of an affected (ff) child = **¼ (25%)**, for *each* pregnancy.

WORKED EXAMPLE

Codominance

Red cattle (C^RC^R) × white (C^WC^W) → all **roan** (C^RC^W). Roan × roan → **1 red : 2 roan : 1 white**.

Always lay out: **parental phenotypes → genotypes → gametes → Punnett square → offspring genotypes → phenotypes and ratio / probability**. Choose letters whose capitals look different from the lower case (F/f, not S/s or C/c).

Reading pedigree diagrams

- Squares = male, circles = female; shaded = affected; horizontal line = partners; vertical line down to children; generations numbered I, II, III.
- **Two unaffected parents with an affected child → the condition is recessive**, and both parents are **carriers** (heterozygous).
- Two affected parents with an unaffected child → dominant.
- **Start with the affected (homozygous recessive) people**, whose genotypes you know for certain, and work outwards.
- An unaffected child of two carriers has a **2/3** chance of being a carrier (FF : Ff = 1 : 2 among the unaffected).

Sex linkage

- Sex is determined by the sex chromosomes: female **XX**, male **XY**. The **Y is much smaller** and lacks many genes found on the X.
- A **sex-linked** gene is carried on the **X chromosome**. Males have only **one X**, so a single recessive allele is **expressed** in a male. Males can't be carriers.
- Fathers pass their X only to **daughters**, so a son can't inherit an X-linked condition from his father.
- **Red-green colour blindness**: recessive allele $\mathbf{X^b}$; normal $\mathbf{X^B}$.

WORKED EXAMPLE

Carrier mother × normal father

Parents: X^BX^b × X^BY · Gametes: X^B, X^b and X^B, Y

	X^B	Y
X^B	X^BX^B normal girl	X^BY normal boy
X^b	X^BX^b carrier girl	X^bY **colour-blind boy**

Probability of a colour-blind child = ¼. Probability a **son** is colour-blind = ½. No daughters are affected (but half are carriers).

MARK-SCHEME WORDING

Genotype: "the **combination of alleles**" (not "genes"). Phenotype: "**observable characteristic**".

Codominance: "**both alleles are expressed**" in the heterozygote, whose phenotype differs from both homozygotes. Not "both dominant".

Genetic diagram marks: parental genotypes → gametes → offspring genotypes → **phenotypes linked to genotypes** → ratio or number. The phenotype mark is the one most often lost. Label ratios ("3 orange : 1 white"), and give **numbers** if numbers are asked for: 284 offspring in a 1 : 2 : 1 ratio → 71 : 142 : 71.

Sex linkage: allele "**on the X chromosome**", recessive; "males need only **one** copy, females need **two**"; "an affected father can't pass it to his sons, because they get his **Y**".

Pedigrees: one reasoned sentence per individual, e.g. "**dominant** because 1 and 2 both have the trait but have a child without it"; "**not X-linked** because female 6 shows the recessive trait but her father doesn't".

Probability of an affected child **of a given sex** = probability affected × ½, e.g. Ff × Ff → affected boy = ¼ × ½ = ⅛.

Often asked as: genetic diagrams; probability MCQs; prove from a pedigree that a condition is dominant / recessive, or not X-linked; sex-linked crosses in new contexts (haemophilia, fruit flies).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **ABO blood groups**: three alleles (I^A, I^B, I^O). I^A and I^B are **codominant**; I^O is recessive. I^AI^B = group AB; I^OI^O = group O. Used in "which baby belongs to which parents" questions.
- **Test cross**: cross an individual with the dominant phenotype with a **homozygous recessive**. All dominant offspring → it was homozygous; about half recessive offspring → it was heterozygous.
- **Haemophilia** is another **X-linked recessive** condition (a missing clotting factor, so blood doesn't clot properly: link to 1.11). It is set up and solved exactly like colour blindness, and explains why far more males are affected.
- Recessive alleles often *are* expressed (they make a non-functional protein), but one dominant allele makes enough working protein, so the phenotype is normal.

- **ZW sex determination** (birds, Komodo dragons): the **female** is ZW (heterogametic) and the male ZZ (homogametic), the reverse of humans (WBI11 Jan 2026).
- **Chi-squared test** to see if offspring numbers fit an expected ratio: work out $(O - E)^2/E$ for each class, add them up, and compare with the critical value at p = 0.05 (WBI11 Oct 2021).

COMMON MISTAKE

Writing sex-linked genotypes without the X and Y (always write X^B, X^b, Y), or giving the Y an allele. Also read the question carefully: "probability that a *child* is affected" differs from "probability that a *boy* is affected".

"Recessive **gene**" or "recessive **disease**": it's a recessive **allele**.

Using X and Y symbols for an **autosomal** cross: it scores 0 or is capped.

Giving the genotype ratio when the phenotype ratio was asked for, or an unlabelled "1 : 3". Forgetting that females can be affected by X-linked conditions.

SPEC 2.16 Cystic fibrosis

WHAT THIS SPEC POINT ASKS

Understand how a gene mutation in cystic fibrosis affects the gas exchange, digestive and reproductive systems.

The cause

- CF is caused by a **recessive allele** of the **CFTR gene**. CFTR is a **chloride ion channel** protein in the membranes of epithelial cells.
- The commonest mutation, **ΔF508**, is a **deletion of three bases** (one codon), so one amino acid (**phenylalanine**) is missing. The protein **folds wrongly** and doesn't work, or isn't inserted in the membrane.

From faulty protein to sticky mucus (learn this chain)

1. Normally, CFTR lets **Cl^- ions** move **out** of epithelial cells into the mucus; Na^+ follows; this **lowers the water potential** of the mucus, so **water moves out** of the cells **by osmosis**, keeping the mucus runny.
2. In CF, Cl^- isn't moved out of the cells, and **Na^+ ions are absorbed** into the cells from the mucus. The water potential **inside the cells** falls below that of the mucus, so **water moves out of the mucus into the cells by osmosis** → the mucus is **dehydrated**: **thick and sticky**.

Effects on the three systems

System	What goes wrong	Consequence
Gas exchange	sticky mucus blocks **bronchioles**; **cilia can't move** it	fewer alveoli ventilated → **smaller surface area** and **longer diffusion distance** → less O_2 uptake (breathlessness); bacteria trapped in mucus → frequent **lung infections**; coughing
Digestive	mucus blocks the **pancreatic duct**; mucus coats the small intestine lining	**digestive enzymes can't reach the small intestine** → food not fully digested; pancreas damaged (cysts, possibly diabetes); thick mucus reduces **absorption** → malnutrition, difficulty gaining weight
Reproductive	thick mucus blocks the **sperm ducts (vas deferens)** in men, or they never form; thick **cervical mucus** in women, and mucus may block the oviducts	men usually **infertile**; women's fertility reduced (sperm can't pass through the cervix to reach the egg)

Inheritance: two carrier parents (Ff × Ff) → **1 in 4** chance each child has CF.

EXAM TIP

Examiners reward the **full chain**: Cl^- not transported → water potential → osmosis → thick mucus → *then* the effect on the specific system. Link gas exchange answers back to **Fick's law** (2.1).

MARK-SCHEME WORDING

The chain: "mutation in the CFTR **gene**" → "CFTR (channel) protein not functioning" → "**chloride ions do not move out** of the cells" → Na^+ moves in → water potential inside the cells falls → "**water leaves the mucus**" by osmosis → "mucus is **dehydrated**", thick and sticky. "Water concentration" is ignored.

Lungs: "**cilia cannot move** the mucus away"; mucus "**traps bacteria**" → infections, coughing, tissue damage; blocked airways → lower concentration gradient, smaller surface area, longer distance → **slower diffusion**.

Digestion: mucus **blocks the pancreatic duct** → enzymes don't reach the small intestine → food not fully digested or absorbed.

Reproduction: "mucus **prevents sperm passing through the cervix**"; mucus may block the oviduct.

ΔF508: "**deletion** removing the **three bases / codon** that code for phenylalanine". **Not accepted**: insertion, or a stop codon.

Often asked as: explain why the mucus is sticky; explain the effect on one system; interpret drug-trial data (FEV1); CF probabilities from carrier parents.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Sweat test**: CFTR also moves Cl^- in sweat glands, so people with CF have salty sweat. Different CFTR mutations leave different amounts of working protein, so chloride levels and symptoms vary (WBI11 Jun 2021).
- **Drug trials**: lung function is measured as **FEV1** (volume breathed out in the first second) with a spirometer; compare the **change in mean** FEV1 (WBI11 Jan 2023).
- A carrier man's sperm: meiosis shares the alleles equally, so **50%** of his sperm carry the faulty allele (WBI11 Oct 2025).

COMMON MISTAKE

"The CFTR **protein** is mutated." The **gene** is mutated; the protein it codes for is faulty.

Writing a generic CF essay: use the diagram or data in the question.

SPEC 2.17 Genetic screening

WHAT THIS SPEC POINT ASKS

(i) Understand the uses of genetic screening: identifying carriers, pre-implantation genetic diagnosis (PGD), and prenatal testing, including amniocentesis and chorionic villus sampling. (ii) Understand the implications of prenatal genetic screening.

Uses

- **Carrier screening**: tests adults (blood or cheek cells) with a family history of a recessive condition, to see if they carry the allele. Couples can make informed decisions about having children.
- **Pre-implantation genetic diagnosis (PGD)**: during **IVF**, one or two cells are removed from each early embryo and tested; only embryos **without** the allele are implanted. Avoids a later termination, but is expensive and unaffected embryos may be discarded.
- **Prenatal testing**: tests the fetus during pregnancy.

Amniocentesis vs CVS

	Amniocentesis	Chorionic villus sampling (CVS)
Sample	**amniotic fluid** containing fetal cells	cells from the **chorionic villi** (fetal part of the placenta)
How	needle through the abdomen, guided by **ultrasound**	needle through abdomen, or tube through the cervix, with ultrasound
When	about **15–20 weeks**	earlier: about **10–14 weeks**
Miscarriage risk	about **0.5–1%**	about **1–2%**
Results	slower (cells may be cultured: 2–3 weeks)	faster

CVS gives an **earlier** result, so any decision (e.g. about termination) is made earlier, but carries a slightly **higher** risk. Results can occasionally be wrong (**false positives/negatives**). (Non-invasive testing of fetal DNA in the mother's blood avoids miscarriage risk.)

Implications of prenatal screening

- **Positive**: parents can prepare (medically, emotionally, financially), choose whether to continue the pregnancy, and the child can be treated early.
- **Negative**: risk of **miscarriage** from the procedure; difficult decisions about **termination**; false results; anxiety; pressure from others.

MARK-SCHEME WORDING

Carrier screening: identifies **heterozygous** couples so they can make an informed decision, but "informed choice" alone isn't enough: **give the options**: no children or adoption; IVF with PGD so "only healthy embryos are implanted"; prenatal testing, then possibly termination. Result: fewer homozygous births.

Amniocentesis: "**amniotic fluid** from around the **fetus**" → fetal cells "analysed for the presence of the **faulty allele**". CVS: "cells / tissue from the **placenta**" ("chronic" villus is rejected).

Risks: "**increases the risk of miscarriage**" (name the procedure); false positives / negatives.

Reasons for a prenatal test: family history, to prepare, to choose an unaffected embryo. Abortion is **not accepted** as the reason.

Often asked as: explain how carrier screening reduces CF births; describe amniocentesis or CVS; 6-mark questions weighing up the options for a named disorder.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Other prenatal tests and timings** (WBI11 Jan 2025): **NIPT** (fetal DNA in the mother's blood, no miscarriage risk) from about 7–10 weeks; CVS; amniocentesis 15–20 weeks; ultrasound; **cordocentesis** (fetal blood from the umbilical cord) at about 22–24 weeks.
- PGD can also screen embryos for the **wrong number of chromosomes** (aneuploidy); it may give false hope and wastes embryos (WBI11 Oct 2021).

COMMON MISTAKE

Treating carrier screening (of adults) as prenatal screening, or confusing PGD (embryos in IVF) with CVS / amniocentesis (a fetus in the uterus).

Use the right word: **embryo** for PGD, **fetus** for CVS and amniocentesis, never "baby".

In 6-mark answers, take each option in turn, use the data given, and tie it to the named disorder. Offering gene therapy is not screening.

SPEC 2.18 Ethical and social issues of genetic screening

WHAT THIS SPEC POINT ASKS

Identify and discuss the ethical and social issues of genetic screening from a range of ethical viewpoints, including religious, moral and social ones.

Frameworks to use

- **Rights-based / deontological**: some actions are right or wrong in themselves (e.g. "every embryo has a right to life", or "people have a right to know or not to know").
- **Utilitarian**: the right choice gives the **greatest good for the greatest number**, weighing suffering prevented against harms.
- **Autonomy**: individuals should make informed decisions about their own bodies and families.

Arguments

For screening	Against / concerns
prevents suffering from a serious genetic disorder (utilitarian)	**religious** belief that life begins at conception: termination and discarding embryos are wrong

For screening	Against / concerns
gives parents information and **choice** (autonomy)	miscarriage risk from amniocentesis/CVS could kill a healthy fetus
lets parents prepare; early treatment	could lead to "**designer babies**" (selecting for sex or non-medical traits)
reduces long-term care costs for health services	**discrimination**: implies that disabled lives are less valuable; insurers or employers might use results
PGD avoids late termination	results can be wrong (false positives/negatives); anxiety; the right **not** to know
	who decides which conditions are "serious enough"?

Genetic counselling before and after testing explains the risks, the meaning of results and the options, without directing the decision.

EXAM TIP

Balanced answers score highest: give **both sides**, name the **viewpoint** each argument comes from (religious, moral, social, economic), and relate it to the **specific technique** in the question (e.g. PGD discards embryos; amniocentesis risks miscarriage).

MARK-SCHEME WORDING

Credited: termination is "**taking a human life**", which is unethical or against religious / cultural beliefs; the embryo "has a **right to life**" and "**cannot give consent**"; "**spare embryos** from IVF are destroyed"; relatives who share the genes "may be **exposed to unwanted facts**"; **false** positives / negatives; miscarriage risk **tied to CVS or amniocentesis**; the perceived worth of an affected child.

Positives: informed choice, time to prepare, fewer people affected.

Not credited: a bare "designer babies" (explain the selection of non-medical traits in IVF instead); "violation of privacy" (too vague).

Often asked as: discuss the ethics of a named test; ethical points inside 6-mark screening questions.

COMMON MISTAKE

One-sided answers: "discuss" means both sides.

Mentioning miscarriage without saying which procedure causes the risk.

Calling an IVF embryo a "fetus" or "baby".

Quick checks

QUICK CHECK

A characteristic shows continuous variation and a bell-shaped distribution. Which explanation is most likely?

- It is controlled by a single gene with two alleles and no environmental effect.
- It is controlled by many genes at different loci and is also affected by the environment.
- It is caused by a single mutation that changes the amino acid sequence of one protein.
- It is controlled by a single locus, and the phenotypes fall into separate groups.

Continuous variation with a normal distribution is produced by polygenic inheritance, where several loci each contribute a small effect, and the environment adds further variation. A single gene with no environmental effect gives discontinuous variation instead, which is the most common wrong answer.

QUICK CHECK

Which change would most directly switch a gene off by making its DNA less accessible to transcription?

- Acetylation of histones at the gene's region
- Demethylation of the cytosine bases in the gene
- Methylation of cytosine bases in the gene's DNA
- Splicing out an intron from the pre-mRNA

Methylation of cytosine next to guanine is linked to gene silencing, because the methyl group reduces transcription. Acetylation and demethylation both tend to activate genes. Splicing acts after transcription and does not change DNA accessibility, so it is the commonest wrong answer.

QUICK CHECK

A single gene produces three different proteins in three different cell types. What is the most likely reason?

- The gene sequence has mutated differently in each cell type.
- Each cell type has a different number of copies of the gene.
- Different combinations of exons are joined during splicing, giving different mRNAs.
- Each protein is produced by a different chromosome that carries the gene.

Alternative splicing joins the exons in different combinations, giving mRNAs of different length and sequence, and therefore different proteins. The DNA is the same in every cell, so mutation is not the explanation. Gene copy number is normally the same in each diploid cell.